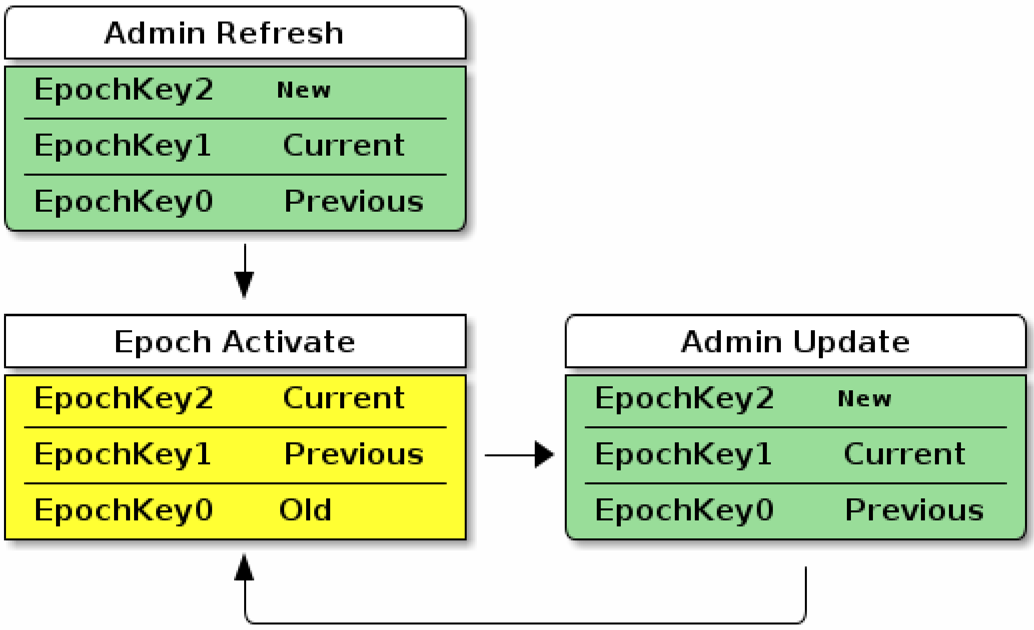

Figure 18. Epoch Key Rotation

The `Admin Refresh` state begins when an entire *group key set* is freshly written to a Node during commissioning or administration, such as when a new group is added. The `Epoch Activate` state is entered when time progresses to activate a fresh *current epoch key*, aging out the other epoch key slots. The `Admin Update` state is entered when an Administrator updates an *old epoch key* with a *new epoch key*. When in steady state, the `Admin Refresh` state MAY be entered in place of an `Admin Update` state transition to update additional keys to the required ones or to completely reset the group security.

Note that in the above diagram, only an example key distribution scheme is illustrated. It is also possible to devise key distribution algorithms that to not rely on time synchronization, or group configurations that never rotate keys in favor or configuring new groups and removing groups and group key sets with expired keys.

Group Key Set ID

The Group Key Set ID is a 16-bit field that uniquely identifies the set of epoch keys used for derivation of an operational group key. Each Group Key Set ID is scoped to a particular Fabric and assigned by an Administrator so as to be unique within a Fabric.

The Group Key Set ID of `0` SHALL be reserved for managing the Identity Protection Key (IPK) on a given Fabric. It SHALL NOT be possible to remove the IPK Key Set if it exists.

4.17.3.6. Group Session ID

A *Group Session ID* is a special case of a 16-bit Session ID that is specifically used for group communication. When Session Type is 1, denoting a group session, the Session ID SHALL be a *Group Session ID* as defined here. The *Group Session ID* identifies probable operational group keys across a Fabric. The *Group Session ID* for a given *operational group key* is derived by treating the output of a Crypto_KDF against the associated Operational Group Key as a big-endian representation of a 16-bit

